

Exploratory Data Analysis & Visualization Framework

AI-Based Pothole Detection and Road Condition Analysis Using Python

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Program	B.Tech — AI & Data Science
Deliverable	Week 2: EDA and Visualization Framework Design
Status	Proposed plan; no dataset analyzed yet

1. Introduction and Purpose

Exploratory Data Analysis (EDA) examines the structure, quality, distributions and relationships within data before model development. This report designs a reusable EDA and visualization framework for a proposed pothole-detection system. The purpose is to plan what should be checked, how findings should be displayed and how the results should guide later data preparation and model training. No real dataset, results or model performance are claimed in this planning document.

2. Proposed Dataset and Data Dictionary

A future dataset would contain appropriately licensed road photographs, with bounding-box annotations for visible potholes where available. The proposed metadata table would have one row per image or annotation. Fields are illustrative, not collected measurements.

Potential field	Type	Use
image_id	Text	Unique image reference
image_width / image_height	Integer	Resolution and aspect-ratio checks
file_size_kb	Numeric	Detect unusually large or small files
pothole_count	Integer	Per-image detection/annotation distribution
bbox_width / bbox_height	Numeric	Object size distribution
bbox_area_ratio	Numeric	Object area divided by image area
lighting / weather	Category (if labeled)	Inspect dataset diversity
source / split	Category	Leakage checks across training/validation/test

3. EDA Questions and Exploration Methods

Approach	Planned checks
Univariate	Image dimensions, file size, pothole count, bounding-box area and lighting-category frequencies.
Bivariate	Image resolution versus pothole count; lighting versus object size; source versus annotation count.
Multivariate	Compare image source, lighting, image resolution and pothole frequency jointly; inspect correlations

	only among meaningful numeric features.
Image-level review	Create representative image grids showing small/large potholes, difficult lighting and potential annotation errors.

4. Data Quality and Cleaning Strategy

- Verify file readability, image format, image dimensions and consistent annotation schema.
- Identify exact duplicates with file hashes; review near-duplicates with perceptual hashes before removing them.
- Check missing image files, missing metadata and missing annotations separately; an image with no potholes may be a valid negative sample.
- Validate bounding-box coordinates (inside image bounds, positive width/height) and flag unusually small or large boxes for manual review.
- Investigate outliers using box plots and robust summaries; do not automatically discard rare but valid road conditions.
- Prevent leakage by grouping related images or adjacent video frames before train/validation/test splitting.
- Record each cleaning decision and the number of affected files in a data-quality log.

5. Proposed Visualizations

Chart / visual	Purpose	Possible tool
Bar chart	Images by source, lighting or annotation category	Seaborn / Plotly
Histogram	Image widths, pothole counts and box area ratios	Matplotlib
Box plot	Compare bounding-box sizes by lighting or source	Seaborn
Scatter plot	Explore resolution versus object size/count	Matplotlib / Plotly
Correlation heatmap	Inspect numeric feature relationships; avoid meaningless encoded categories	Seaborn
Image contact sheet	Inspect samples and annotation overlays visually	OpenCV / Matplotlib
Missingness matrix	Show metadata completeness where applicable	Pandas / Seaborn

6. Illustrative Visualization Workflow

Inventory images and metadata → Validate files and labels → Generate summary statistics → Review duplicates, missing values and outliers → Create univariate plots → Compare selected variable pairs → Inspect representative annotated images → Record findings and recommended next steps.

7. Python Tools and Their Roles

- Pandas: metadata tables, grouping, null-value audits and descriptive summaries.
- NumPy: numerical calculations, quantiles and derived bounding-box features.
- OpenCV / Pillow: image dimensions, file integrity checks, resizing previews and drawing boxes.

- Matplotlib: static histograms, scatter plots and image grids.
- Seaborn: box plots, category comparisons and correlation heatmaps.
- Plotly: optional interactive filters and exploratory charts.
- Jupyter Notebook: reproducible narrative, code, charts and decisions.

8. Documentation and Reporting Framework

The later EDA report will contain: (1) dataset provenance and licensing; (2) dataset size and schema; (3) missingness, duplicates and annotation-quality summary; (4) descriptive statistics; (5) plots with clear captions and axis labels; (6) representative image examples; (7) observed limitations and potential bias; and (8) recommendations for preprocessing, augmentation and model evaluation. Charts will distinguish measured results from illustrative examples and cite the actual dataset once selected.

9. Work Plan — 35 Hours

Task	Hours
Define EDA questions and dataset schema	4
Plan image/annotation validation	5
Design cleaning, duplicate and outlier checks	5
Design univariate and bivariate analysis	6
Plan visualization examples and layouts	6
Select Python libraries and notebook structure	3
Write reporting and interpretation template	3
Review, refine and document framework	3
Total	35

10. Expected Outcomes and Limitations

Expected deliverables are an actionable EDA checklist, proposed data dictionary, chart selection matrix, reproducible analysis workflow and reporting structure. These are plans, not empirical findings. Actual plots, statistical observations and dataset-specific recommendations require a real, appropriately licensed dataset in a later stage.

11. Risks and Mitigation

Risk	Planned mitigation
Unbalanced road conditions	Stratify exploratory summaries by source, weather and lighting when metadata exists.
Annotation inconsistency	Define labeling rules and review a sample of bounding boxes.
Data leakage	Group related frames and near-duplicate images before splitting.
Misleading correlations	Review distributions and data types; interpret associations cautiously.
Privacy / licensing	Use permitted images; avoid exposing faces or number plates in examples.

12. Conclusion

This Week 2 framework defines how a future pothole-image dataset would be explored, cleaned, visualized and reported using Python. It prepares the project for informed preprocessing and object-detection model development without claiming that the analysis has already been performed.

Reference Documentation

- Pandas: <https://pandas.pydata.org/docs/>
- Matplotlib: <https://matplotlib.org/stable/>
- Seaborn: <https://seaborn.pydata.org/>
- Plotly: <https://plotly.com/python/>
- OpenCV: <https://docs.opencv.org/>